Lecture 22: Autoencoders

A network that learns to copy - and why the copy is the point

From 784 numbers to 2, and back

One MNIST digit is a vector of **784** numbers (28×28 pixels). Could you describe it with just **2**? Think of describing a *person* by their age and height - lossy, but wonderfully compact.

Predict: can a network learn that squeeze *and* rebuild the digit from the 2 numbers - with *no labels*, ever?

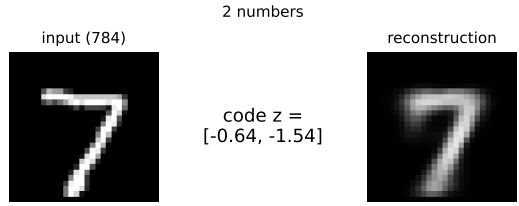
From 784 numbers to 2, and back

One MNIST digit is a vector of **784** numbers (28×28 pixels). Could you describe it with just **2**? Think of describing a *person* by their age and height - lossy, but wonderfully compact.

Predict: can a network learn that squeeze *and* rebuild the digit from the 2 numbers - with *no labels*, ever?

Basically yes (right). The trick: the network's **target is its own input**. No human labels - it **supervises itself**.

Chapters 5-7 (nets, CNNs, RNNs) were *supervised*. Autoencoders take us back to **unsupervised** learning - the territory of clustering (ch4) and PCA (ch4b) - now with a neural net and no y.



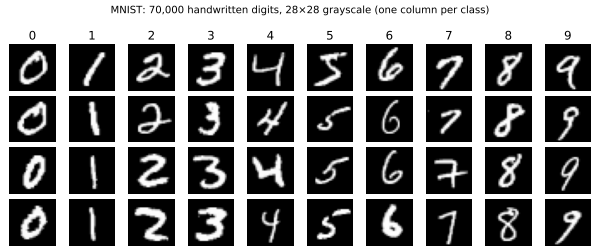
Meet the data: MNIST

Our running dataset: **MNIST** - 70,000 handwritten digits, each 28×28 grayscale ($= 784$ numbers).

- **LeCun, Cortes & Burges** (1998), from US Census / high-school forms.
- The *"hello world" of computer vision* - the first benchmark for any new method.
- Small, clean, CPU-friendly.

An autoencoder only looks at the **pixels**; the labels (0-9) are just for the pictures.

And not just images: an autoencoder works on *any* vector - tabular rows, sensor time-series, word or audio embeddings.



Outline

Why care? Autoencoders are everywhere

Before we build them: the ideas in this short chapter are load-bearing in today's biggest models.

- **Self-supervised pretraining. BERT** masks words and rebuilds them; **MAE** masks image patches - "masked modeling" is an autoencoder idea (denoising), and the recipe behind most foundation models.
- **LLM interpretability.** A wide **sparse autoencoder** on a model's activations splits them into single-concept features - how Anthropic reads what is inside Claude (*Scaling Monosemanticity*, 2024).
- **Cheaper inference. DeepSeek**'s Multi-head Latent Attention squeezes the KV cache into a small **latent vector** ($\sim$4% of the size), so long contexts fit in memory (DeepSeek-V3, 2024).
- **Image generation.** The **VAE** (next lecture) is the compression layer inside Stable Diffusion and Midjourney.

Encoder, bottleneck, denoising, sparse - we build each piece below; watch for where it powers the models above.

The autoencoder idea

Learn to rebuild your input, and let the bottleneck do the teaching.

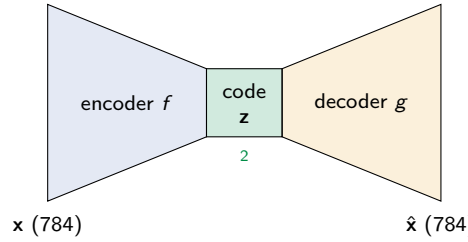
Encoder, code, decoder

An **autoencoder** is two networks trained together:

- **Encoder** f : $\mathbf{x} \mapsto \mathbf{z}$ - compress the input into a small **code z**.
- **Decoder** g : $\mathbf{z} \mapsto \hat{\mathbf{x}}$ - rebuild the input from that code.

Train so $\hat{\mathbf{x}} \approx \mathbf{x}$; the target is the input itself, so it needs **no labels** (self-supervised).

Code / latent. The code **z** is a compact summary of the input; the space it lives in is the **latent space** ("latent" = hidden / underlying). We call **z** a *code* because the encoder **encodes x** into it - a short "password" that stands for the whole image.



The bottleneck is the whole point

Make the code **smaller** than the input: $\dim(\mathbf{z}) < \dim(\mathbf{x})$ (**undercomplete**). The network cannot pass everything through, so it must keep only what matters - the structure the digits share.

No bottleneck, no learning. If $\dim(\mathbf{z}) \geq \dim(\mathbf{x})$ the network can just learn the **identity** - a perfect, useless copy that memorized nothing.

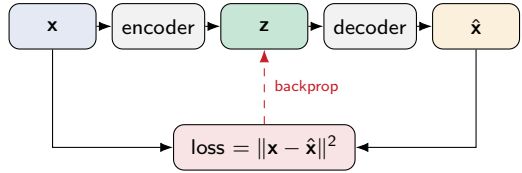
The **loss** is nothing new: squared error $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$ (or cross-entropy for $[0, 1]$ pixels). *Callback:* the L2 loss from regression.

How an autoencoder learns

It is the same training loop as any net (ch5, L15) - only the *target* is unusual:

1. **Forward:** encode $\mathbf{z} = f(\mathbf{x})$, then decode $\hat{\mathbf{x}} = g(\mathbf{z})$.
2. **Loss:** compare the output to the **input itself**, $L = \|\mathbf{x} - \hat{\mathbf{x}}\|^2$.
3. **Backward:** backprop L through decoder *and* encoder.
4. **Update:** one Adam step; repeat over mini-batches.

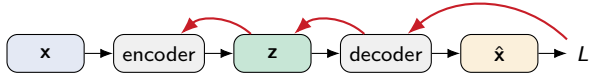
The one twist vs a normal net: the label *is* the input, so there is nothing to hand-label. Everything else - forward, backprop, Adam - is exactly ch5.



Backprop, in symbols

$\hat{\mathbf{x}} = g\big(f(\mathbf{x})\big)$ is a **composition**, so the chain rule sends the error $L = \|\mathbf{x} - \hat{\mathbf{x}}\|^2$ back through **both** halves (θ_f, θ_g = encoder / decoder weights):

$$\underbrace{\frac{\partial L}{\partial \theta_g}}_{\text{decoder}} = \frac{\partial L}{\partial \hat{\mathbf{x}}}\,\frac{\partial \hat{\mathbf{x}}}{\partial \theta_g}, \qquad \underbrace{\frac{\partial L}{\partial \theta_f}}_{\text{encoder}} = \underbrace{\frac{\partial L}{\partial \hat{\mathbf{x}}}\,\frac{\partial \hat{\mathbf{x}}}{\partial \mathbf{z}}}_{\text{back through the decoder}}\,\frac{\partial \mathbf{z}}{\partial \theta_f}.$$



gradient flows back: through the decoder → the code → the encoder

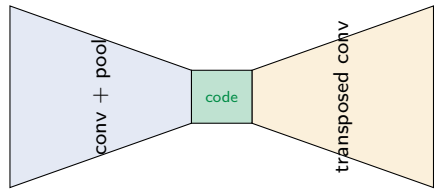
The **encoder** is trained *through* the **decoder**: the decoder first says "this code would rebuild **x** better", then the encoder learns to produce such codes - which is why the two halves share one representation. `loss.backward()` does it all.

You already built the decoder

For images, the two halves are just a **CNN and its mirror** - a **convolutional autoencoder**:

- Encoder = convolution + pooling (*ch6, L16*) - shrink to a code.
- Decoder = **transposed convolution** (*L19*) - grow the code back to an image.

Nothing new to learn here: you built both halves in the CNN chapter. An autoencoder just points them at each other.



In practice (PyTorch)

A working autoencoder is a handful of lines - encoder, decoder, and the copy-yourself loop:

```
enc = nn.Sequential(
  nn.Linear(784,256), nn.ReLU(),
  nn.Linear(256,32))          # code
dec = nn.Sequential(
  nn.Linear(32,256), nn.ReLU(),
  nn.Linear(256,784), nn.Sigmoid())

for xb in loader:             # no labels!
  xh = dec(enc(xb))
  loss = F.mse_loss(xh, xb)
  opt.zero_grad()
  loss.backward(); opt.step()
```

Practical notes

- Scale pixels to $[0, 1]$; use `mse_loss` (or `BCE` with the `Sigmoid` output).
- The code size is your main knob - start around **16-64**.
- **Identity trap**: perfect reconstruction but a useless code $\Rightarrow$ the code is too big. Shrink it, or add noise / an L_1 penalty.
- For images, swap `Linear` for `Conv2d` / `ConvTranspose2d`.

Autoencoders = nonlinear PCA

You already compressed digits once - PCA. This bends the axes.

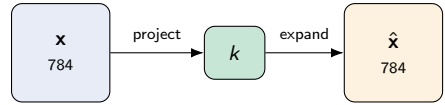
PCA is already an encoder-decoder

Rewind to **PCA** (ch4b). To compress a digit it does two steps:

- **Encode:** project the 784 pixels onto the top k directions $\rightarrow$ k numbers.
- **Decode:** multiply those k numbers back out $\rightarrow$ a reconstruction $\hat{\mathbf{x}}$.

That is *exactly* an autoencoder - only with **linear** encode and decode and no hidden layers.

So PCA is the **simplest possible autoencoder**. The real question: what changes when the encoder and decoder become real (nonlinear) networks?



both arrows are just matrix multiplies

From PCA to a real autoencoder

Keep the encoder and decoder **linear** - the activation is the *identity* $\phi(a) = a$ (just weighted sums, no bend) - and minimize reconstruction error: the autoencoder provably recovers the **same subspace as PCA** - the best flat sheet through the data (directions of maximum variance).

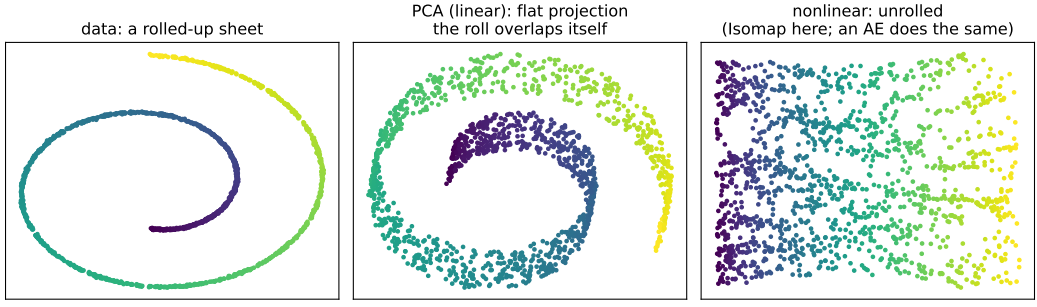
Now switch the activations **on** (ReLU, sigmoid). Encode and decode become **curved**, so the code can follow structure a flat sheet cannot. That is the whole idea:

linear autoencoder = PCA	$\longrightarrow$	**nonlinear** autoencoder = PCA that can bend

You do not throw PCA away - you **generalize** it. The next two slides show the difference: first on a curved toy dataset, then on the digits.

See it: PCA flattens, the autoencoder bends

A **swiss roll** - a 2-D sheet rolled up in 3-D. PCA can only take a flat photo of it; a nonlinear method unrolls it back to the sheet:



In PCA's flat photo (middle) far-apart points on the roll **land on top of each other**; the nonlinear map (right; Isomap here, an autoencoder does the same) **unrolls** it so the colour runs smoothly. That is what "bend" buys you.

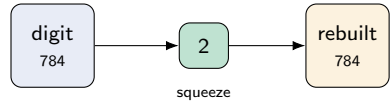
Back to digits: the 2-number experiment

Same task, head to head. Force each test digit through a **2-number** bottleneck, then rebuild it:

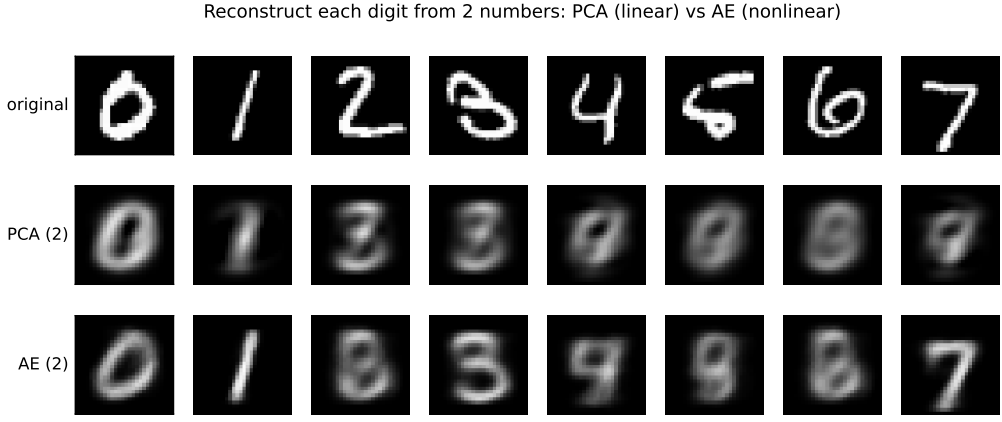
- **PCA**: keep only the top **2 components**, reconstruct (linear).
- **Autoencoder**: a $784 \to 2 \to 784$ net with nonlinear activations.

Both get the *same* 2-number budget; the only difference is linear vs curved.

2 numbers is a brutal squeeze for a 784-pixel digit - neither will be perfect. The question is *which structure survives*.



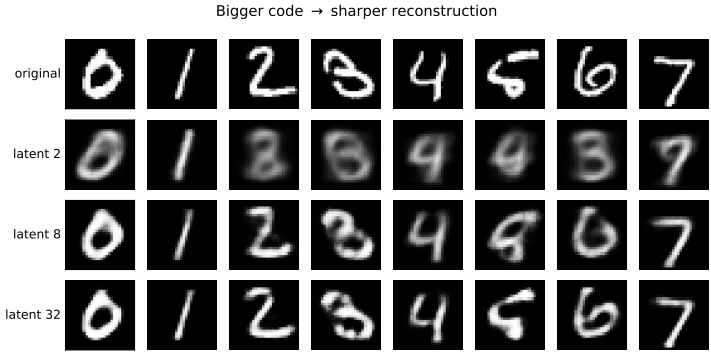
The result: nonlinear keeps more



PCA's 2-component reconstruction (middle row) collapses several digits to a ghostly *average* - 4, 5, 6, 7 all blur into a similar blob. The autoencoder's 2-number code (bottom row) keeps more real digit shape. Same budget - the curve wins.

How big should the code be?

The code size - the **latent dimension** - is the key knob (we used 2 only so we could *draw* the space):

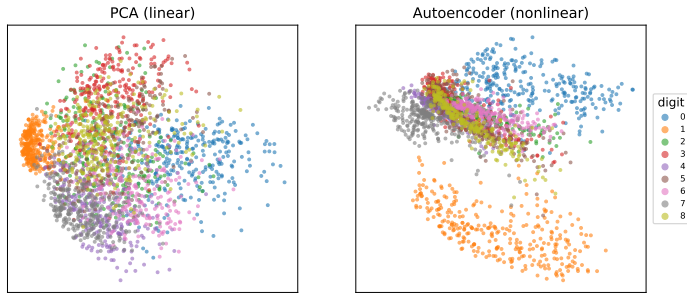


Too small (2): the net **guesses** (a 5 comes back a 9). Bigger (8, 32): sharper, classes separate - but you lose the 2-D picture. Real autoencoders use **tens to hundreds**.

The same digits as a 2-D map

Both methods also give a **2-D map** of the data (each point is one digit).

Honest caveat: the AE's axes are **not** orthogonal, **not** variance-ordered, and **not** unique (another training run gives a different map). PCA hands you all of those for free. The AE trades them for flexibility.



The latent captures more than pixels

That 2-D map is not just pretty - the code captures **structure you can use**.

A striking example from medicine: a patient's **sex** is nearly impossible to read off a raw **brain MRI** - even trained radiologists cannot do it by eye. Yet train an autoencoder on the scans, look at the latent, and **men and women separate**.

The autoencoder surfaced a signal that *was* in the data but hidden from the eye - here enough to fill a missing field in a patient record. That is **representation learning**: a compact code that makes hidden structure explicit. *(Illustrative - a known result on brain imaging, not run here.)*

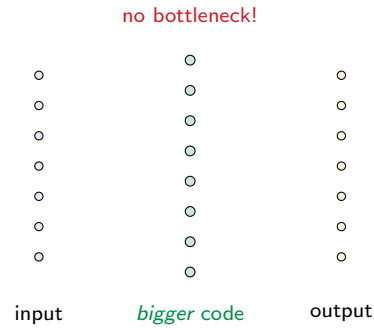
Regularized autoencoders

When you cannot rely on a tight bottleneck, constrain the code another way.

Why go overcomplete?

Compression is not always the goal. Sometimes you want to **re-represent** the input as a *richer* set of features - a code **bigger** than the input, called **overcomplete**. Two reasons it helps:

- **A vocabulary, not a summary.** A language has thousands of words, but any one sentence uses a handful. An overcomplete code is a big **dictionary** of feature-detectors; each image switches on just the **few** it needs (like the brain: many neurons, few firing at once).
- **Spread out to separate.** Structure tangled in the raw pixels can become easy to read once the input is lifted into a bigger, richer space.



...but a big code can cheat

An overcomplete code has room to spare - so the network can take the lazy way out and just **copy the input straight through** (the identity map again), learning nothing useful.

With a big code we can no longer lean on a tight bottleneck to force learning. We need a *different* constraint to keep the network honest. Three classic fixes, coming up: **denoising** (corrupt the input), **sparse** (penalize how much the code fires), and **contractive** (penalize the code's sensitivity to the input).

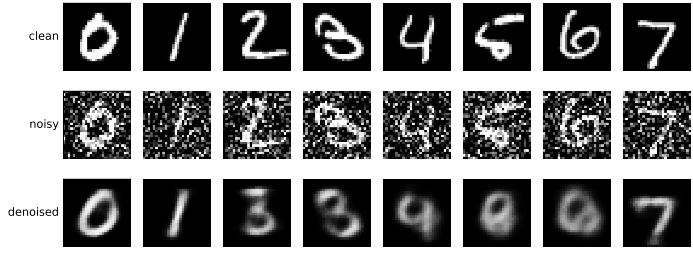
Denoising autoencoders

Break the input on purpose. Corrupt **x** (add noise, or zero out pixels), then train the network to output the **clean** original.

It cannot copy noise, so it must learn what a digit really is.

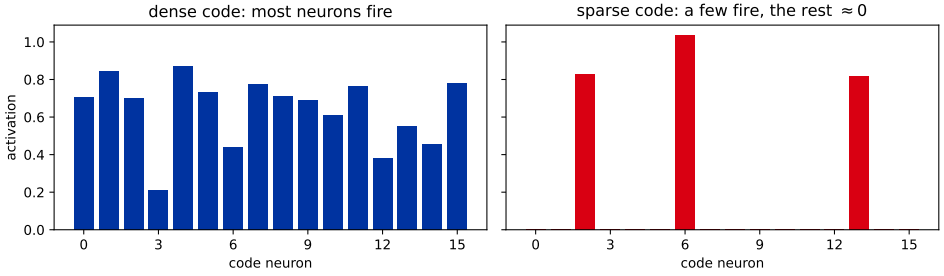
Callback: ch5 data augmentation - robustness by perturbation. Here the perturbation is the whole training signal.

Predict: zero out half the pixels and demand the clean digit back - does that really help? *Yes.*



Sparse autoencoders

Constraint #2: don't shrink the code, **penalize how much it fires**. Add an L_1 penalty on the code activations (or a target average activation), so for any one input only a **few** of the many code neurons switch on.



Each input now lights up a **small, specialised** set of detectors - more interpretable, and it *cannot* copy the input (that would need every neuron). *Callback:* the L_1 penalty is Lasso (ch1), now on **activations** instead of weights.

Contractive autoencoders

Constraint #3: make the code **insensitive to tiny input wiggles**. Penalize how fast the encoder f reacts as the input moves - the squared Frobenius norm of its Jacobian:

$$L(\mathbf{x},\, g(f(\mathbf{x}))) \;+\; \lambda \left\| \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} \right\|_F^2 .$$

Now two nearby inputs map to **almost the same code**: the encoder contracts small perturbations away and keeps only the few directions that actually matter - those along the data manifold.

Denoising vs contractive: the DAE earns a stable code by **corrupting inputs** (sampled noise); the CAE asks for it **directly**, through a deterministic Jacobian penalty. Same goal, two routes. *Callback:* it is just another term added to the loss - exactly like weight decay in ch5.

Anomaly detection

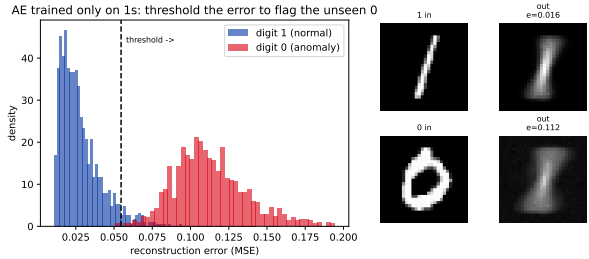
If it cannot rebuild you, you do not belong.

Reconstruction error is an anomaly score

Train an autoencoder on **normal data only**. It learns to rebuild normal inputs with **low** error. Feed it something it has never seen - it rebuilds it **badly**, with **high** error.

So the reconstruction error *is* an anomaly score: pick a **threshold**, flag anything above it.

The industrial workhorse use of autoencoders: fraud, manufacturing defects, network intrusion. *Callback:* thresholding a score - ch3 metrics.



Trained only on 1s; a 0 gets forced toward a 1-shape ⇒ high error.

A caveat you can see

Reconstruction-error anomaly detection only works when the anomaly is **genuinely different** from the training data.

Train on digits 0-8 and try to flag a **9**: it fails - a 9 looks like a 4, a 7, a 1, all of which the network saw, so it reconstructs 9s just fine. The clean split on the previous slide needed a truly distinct anomaly (train on 1s, flag 0s). Know what "normal" and "anomalous" mean for *your* data before you trust the error.

Recap: autoencoders

- An autoencoder is an **encoder + decoder** trained to rebuild its input - **self-supervised**, no labels.
- The **bottleneck** forces it to keep only structure; a linear one recovers **PCA**, a nonlinear one bends around curved data.
- Cannot rely on a tight bottleneck? Regularize: **denoising** (corrupt and recover), **sparse** (penalize activations), or **contractive** (penalize code sensitivity).
- **Anomaly detection**: high reconstruction error flags what the network never learned.

But it cannot generate

One thing the autoencoder **cannot** do: invent a new digit. To generate, you would draw a random code **z** and decode it. But the AE never told its codes to follow any known distribution.

Draw $\mathbf{z} \sim \mathcal{N}(0, I)$ (red) and the codes (blue) are somewhere else entirely: the draws miss almost the whole manifold and decode to blurry blobs. You can **reconstruct**, but you cannot **sample**.

Next (L23): give the latent space no holes, then sample it - **variational autoencoders**.